Section 1.2 — Inequalities and Intervals — Answer Key

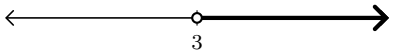
Math 103: College Algebra

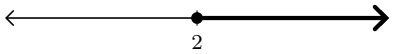
1 Intervals in the Number Line

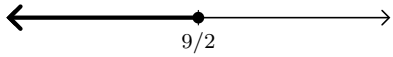
1. a. $x < -4$ b. $(-\infty, -4)$
2. a. $-4 \leq x < 4$ b. $[-4, \infty)$
3. a. $0 \leq x \leq 3$ b. $[0, 3]$
4. a. $x > -2$ b. $(-2, \infty)$
5. a. $x \leq -3$ b. $(-\infty, -3]$
6. a. $x \geq 2$ b. $[2, \infty)$
7. a. $-5 < x \leq -3$ b. $(-5, -3]$
8. a. $-2 < x < 5$ b. $(-2, 5)$

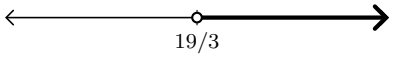
2 Solving Inequalities

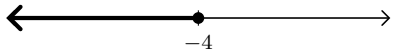
For each inequality below:

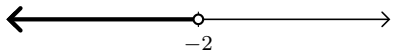
9. a. $x > 3$ b. 

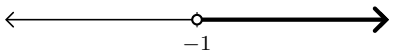
c. $(3, \infty)$
10. a. $x \geq 2$ b. 

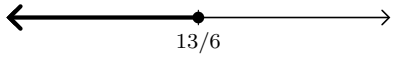
c. $[2, \infty)$
11. a. $x \leq \frac{9}{2}$ b. 

c. $(-\infty, \frac{9}{2}]$
12. a. $x > \frac{19}{3}$ b. 

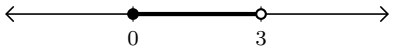
c. $(\frac{19}{3}, \infty)$
13. a. $x \leq -4$ b. 

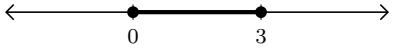
c. $(-\infty, -4]$
14. a. $x < -2$ b. 

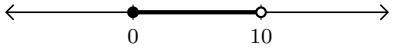
c. $(-\infty, -2)$
15. a. $x > -1$ b. 

c. $(-1, \infty)$
16. a. $x \leq \frac{13}{6}$ b. 

c. $(-\infty, \frac{13}{6}]$
17. a. $1 < x < 2$ b.

c. $(1, 2)$
18. a. $0 \leq x < 3$ b. 

c. $[0, 3)$
19. a. $0 \leq x \leq 3$ b. 

c. $[0, 3]$
20. a. $0 \leq x < 10$ b. 

c. $[0, 10)$

3 Union and Intersection

Including sketches here to make it easier to see how the union and intersection were found.

21. Sketch:

$[-14, 14)$

22. Sketch:

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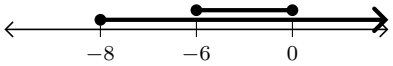
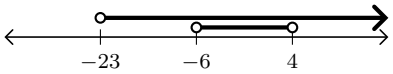
23. Sketch: 

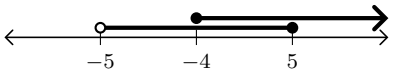
$[-8, \infty)$

24. Sketch: 

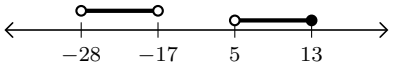
$[-23, \infty)$

25. Sketch:

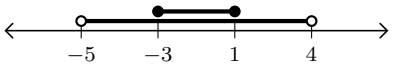
$[-8, 8]$

26. Sketch: 

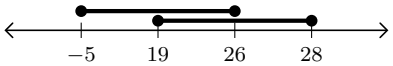
$(-5, \infty)$

27. Sketch: 

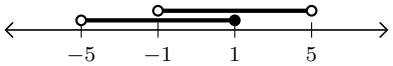
$(-28, -18) \cup (5, 13]$

28. Sketch: 

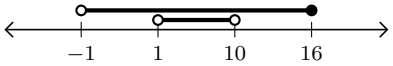
$[-3, 1]$

29. Sketch: 

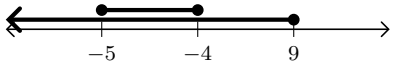
$[-5, 28]$

30. Sketch: 

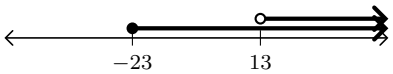
$(-1, 1]$

31. Sketch: 

$(1, 10)$

32. Sketch: 

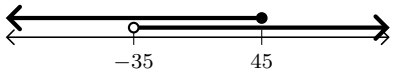
$[-5, -4]$

33. Sketch: 

$(13, \infty)$

34. Sketch:

$[-7, 9]$

35. Sketch: 

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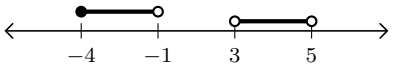
36. Sketch: 

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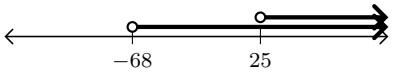
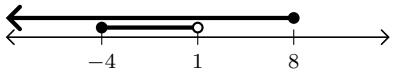
37. Sketch: 

$(-68, \infty)$

38. Sketch: 

$[-4, 1)$